

Causal Inference Crash Course:

Part 10 — Conformal Inference

Julian Hsu

Causal Inference Series

1. Foundations
2. Causal Models
3. Inference
4. Best Practices
5. Heterogeneous Treatment Effects
6. Panel Models
7. Regression Discontinuity
8. Surrogate Models
9. Arguable Validation
10. **Conformal Inference** ← **this deck**

Overview

- **So what?** Every model's point prediction is wrong by some amount — conformal inference is how you say honestly by how much. It turns *any* prediction into a **set** with a **finite-sample, distribution-free coverage guarantee**, needing only **exchangeability**.
- The payoff: this same machinery gives **valid causal inference with a single treated unit** (synthetic control), where classical standard errors go quiet.
- We cover:
 - the idea: **split conformal** and its coverage guarantee
 - the one assumption — **exchangeability** — and what breaks it
 - two fixes when it fails: **moving blocks** and **weighting**
 - the causal payoff: conformal inference for **synthetic controls**
- Assumes the earlier **Inference** and **Panel Models** decks.

The Idea

You don't have to trust the model — you just have to believe the data's story doesn't depend on the order you saw it in.

What we want

- Here's the uncomfortable question nobody asks about a point prediction: how often is it *this* far off? Most workflows don't say — conformal inference answers it honestly, without pretending the model is right.
- Classical prediction intervals need the **model to be right** — Wald wants a correct parametric form, Bayes a correct likelihood. Neither survives contact with the black-box model you actually shipped.
- **Conformal** asks for far less: a set $\mathcal{C}(x)$ with

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

in **finite samples**, for **any** distribution, wrapped around **any** predictor — however ugly.

- Running example we'll carry through: predict a customer's next-quarter purchases, and give an interval that really covers 90% of the time.

The one assumption: exchangeability

- Here's the trade: give up on knowing the model is right, and in exchange you only need to believe the data's underlying pattern doesn't care about order. That's a much easier thing to defend.
- Formally: $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are **exchangeable** — their joint distribution is unchanged by reordering them. (i.i.d. is the familiar special case.)
- That's the *whole* assumption — no distributional form, no correctly specified model. Everything downstream is a consequence of it.
- And it's exactly what **fails** for time series and panels — the moment your data has a "before" and "after," this assumption is in trouble. We'll come back to that.

Any Score Works — a Good One Just Tightens the Set

- Before you can say "how often are we wrong," you need a ruler for "how wrong." That ruler is the **nonconformity score** $s(x, y)$.
- Regression default: the absolute residual of any fitted $\hat{f}$,

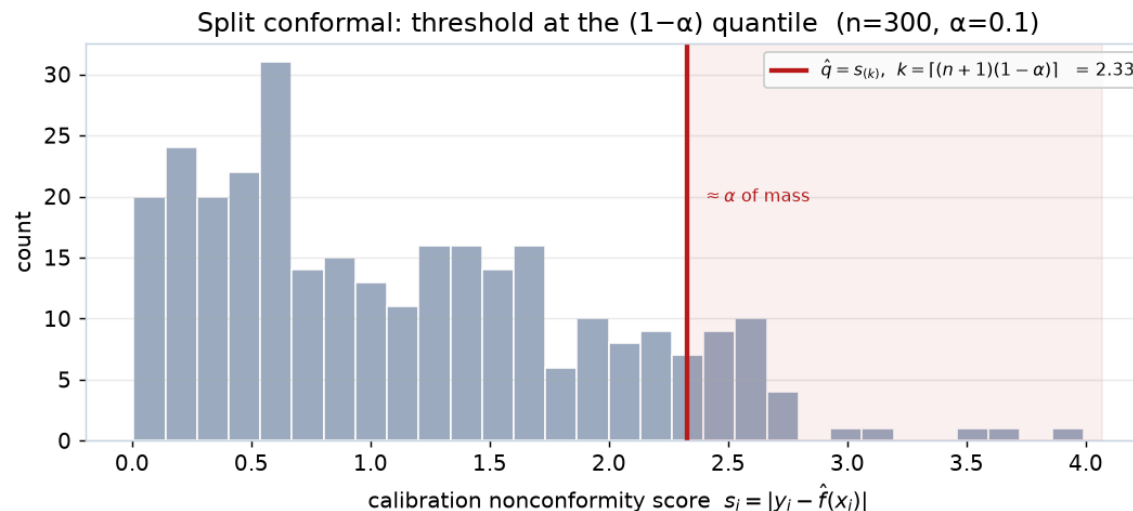
$$s(x, y) = |y - \hat{f}(x)|.$$

- The guarantee holds for **any** score and **any** model — a *good* score doesn't buy validity, you already have that. It only makes the set **tighter**.
- *Details: [Appendix — Choosing a Conformity Score](#)*

Split conformal — the recipe

The mechanics, step by step. For miscoverage level α , split the data into a **training** fold and a **calibration** fold of size n :

1. **Fit** $\hat{f}$ on the training fold only.
2. **Score** the calibration set: $s_i = s(X_i, Y_i)$, $i = 1, \dots, n$.
3. **Quantile**: $\hat{q} = s_{(k)}$, the k -th smallest score, with $k = \lceil (n + 1)(1 - \alpha) \rceil$.
4. **Predict**: $\mathcal{C}(x) = \{y : s(x, y) \leq \hat{q}\}$.



The Guarantee Holds Even When Your Model Is Wrong

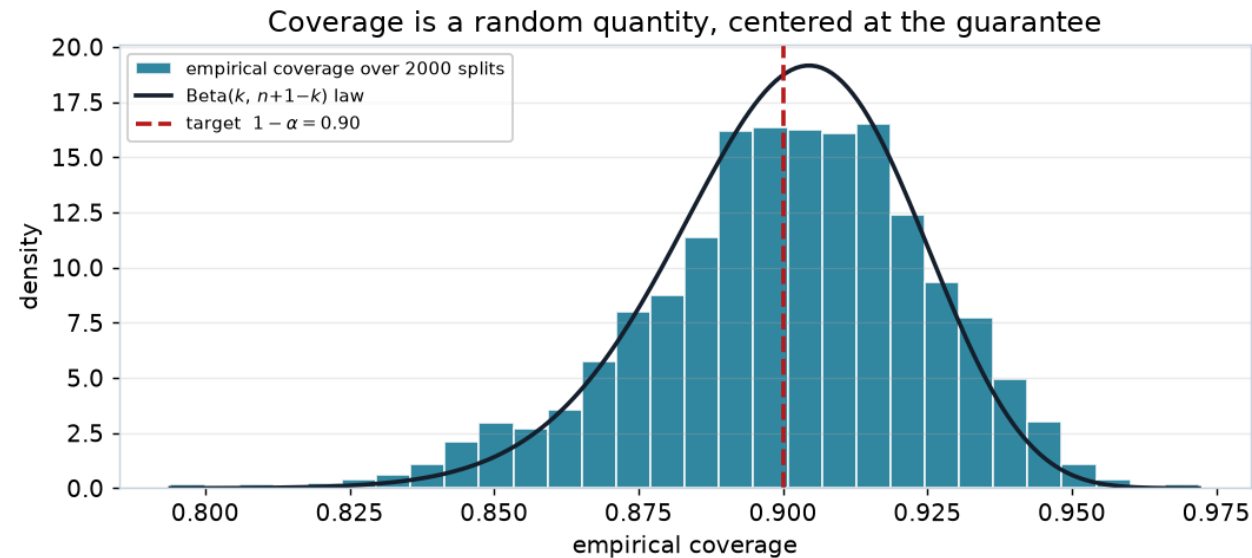
- This is the payoff of the last two slides: a real, provable promise about how often your set is right — not "should be fine in practice."
- **Theorem** (Vovk et al. 2005; Lei et al. 2018). If the data are **exchangeable**, split conformal satisfies

$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}.$$

- **Finite-sample, distribution-free, model-free.** The upper bound needs only that scores have no ties.
- The model can be **arbitrarily wrong** — coverage still holds. A bad model just gets you an honestly **wide** set, not a broken guarantee.
- *Details: [Appendix — Why Coverage Survives a Bad Model](#)*

Why it works

- Insert the unseen test score as an $(n+1)$ -th point. By exchangeability its **rank is uniform** on $\{1, \dots, n+1\}$, so it exceeds the k -th calibration value only about α of the time — exactly the miscoverage event.
- So coverage is a **property of the procedure**, not luck: over repeated draws it follows a $\text{Beta}(k, n+1-k)$ law centered at $1 - \alpha$.



Conformal Prediction Is Hypothesis Testing in Disguise

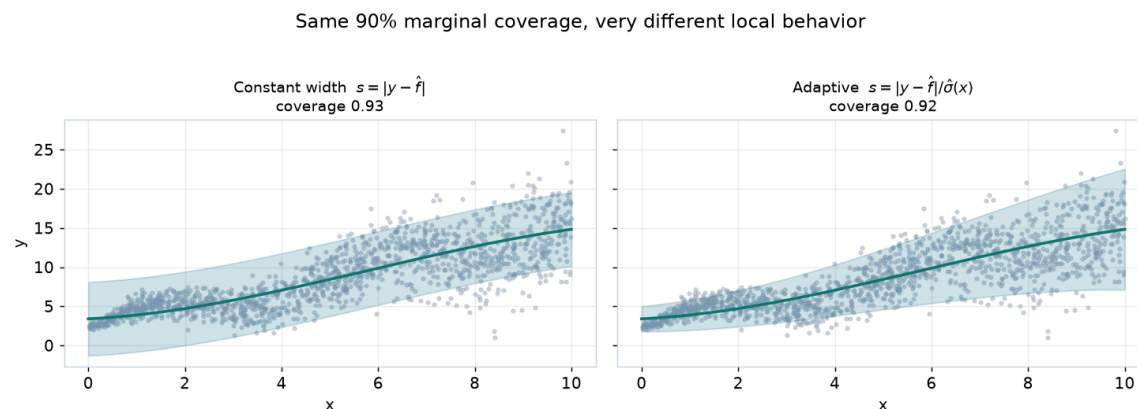
- Equivalent framing (Shafer & Vovk 2008): conformal is **hypothesis testing turned into prediction**. For each candidate label y ,

$$p(y) = \frac{1 + \#\{i : s_i \geq s(x, y)\}}{n + 1}, \quad \mathcal{C}(x) = \{y : p(y) > \alpha\}.$$

- Keep the labels you **cannot reject** at level α .
- This is the view we'll lean on for synthetic control — where the "label" becomes a hypothesized treatment effect, and rejecting it is the whole game.
- *Details: [Appendix — Conformal Sets as Test Inversion](#)*

Adaptivity — and its limit

- **Heads up:** the guarantee we just proved is **marginal**, not conditional — some regions can be **under-covered** and others **over-covered**; exact conditional coverage is impossible distribution-free.
- Fix with a smarter score, not a new guarantee: normalize by a local scale $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$, or use **conformalized quantile regression** (Romano, Patterson & Candès 2019).

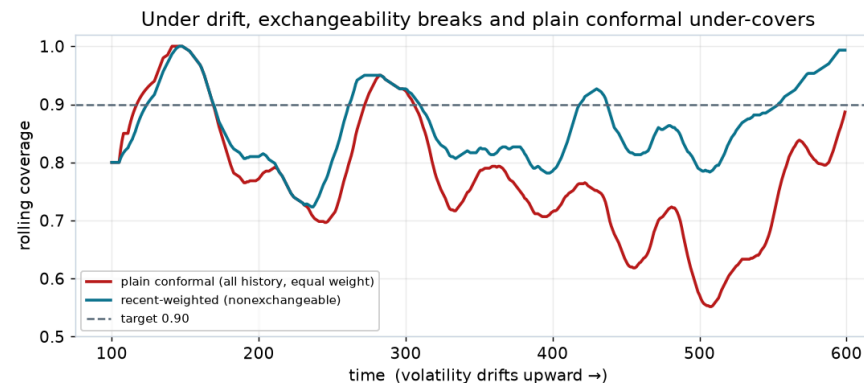


When Exchangeability Fails

Real data has a "before" and "after." That breaks the one assumption we just leaned on — here's what to do about it.

Time and treatment break exchangeability

- Everything so far leaned on one belief: that reordering the data wouldn't change its story. Time series don't grant that — there's a real "before" and "after," and most data we actually work with has one.
- Panels and time series are **serially dependent** and often **drift** — reorder them and the joint law changes, so exchangeability fails.
- Worse for causal work: the null holds only **before** treatment, and the post-period is exactly what we suspect differs.
- Plain conformal then **under-covers** — calibrated on a stale past that no longer resembles the present.



Fix A — permute blocks, not points

- If shuffling the data breaks its story, don't shuffle points — shuffle in chunks big enough to keep the story intact.
- Keep **contiguous blocks** of residuals together and permute the block **positions** — the moving-block idea borrowed from the block bootstrap.
- Local dependence rides along *inside* each block; only its placement is randomized, so the null datasets keep the real time structure.
- **Exact** under exchangeability, **approximately valid** under stationarity and weak dependence — the block length must exceed the dependence range (Chernozhukov, Wüthrich & Zhu 2018, 2021).

Fix B — weight what belongs

- The alternative move: don't chunk the data, just admit some points matter more than others — and say so explicitly, with weights fixed in advance.
- **Nonexchangeable conformal** (Barber, Candès, Ramdas & Tibshirani 2023): attach fixed weights w_i and take **weighted quantiles** of the scores — for time series, weight **recent** points more.
- The guarantee degrades gracefully rather than breaking outright:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha - \Delta,$$

where Δ is a weighted **distance from exchangeability**; it vanishes when the up-weighted points really do resemble the target.

- One rule: the weights must be **fixed a priori** — not learned from the scores, or you're back to cheating.

Two relaxations of one idea

- **Back to the big idea:** both fixes are the same move — restore just enough symmetry for the uniform-rank argument to go through, without pretending the data is something it isn't.
- **Blocks** — the *discrete* relaxation: glue dependent points together so a permutation can't shatter the local structure.
- **Weights** — the *continuous* relaxation: down-weight the points that don't belong with the target.

Conformal Inference for Synthetic Controls

The payoff: turn this same machinery into a causal test that works with just one treated unit.

Causal Inference = Prediction + a Break Test

- Here's the payoff we've been building to: everything up to now was about intervals for predictions. This section turns the same machinery into a **causal test** — with just one treated unit, no fancy standard errors required.
- **Chernozhukov, Wüthrich & Zhu (2021)** glue two familiar problems together: causal inference = **counterfactual prediction** + **a test for a structural break**.
- Predict the treated unit's *untreated* path (synthetic control, DiD, factor models...), then ask whether the **post-treatment residuals break** from the pre-treatment ones.
- The pitch: **inference that doesn't care how good your synthetic control is** — validity comes from the permutation, not from a correct model.
- Details: [Appendix — The CWZ Framework](#)

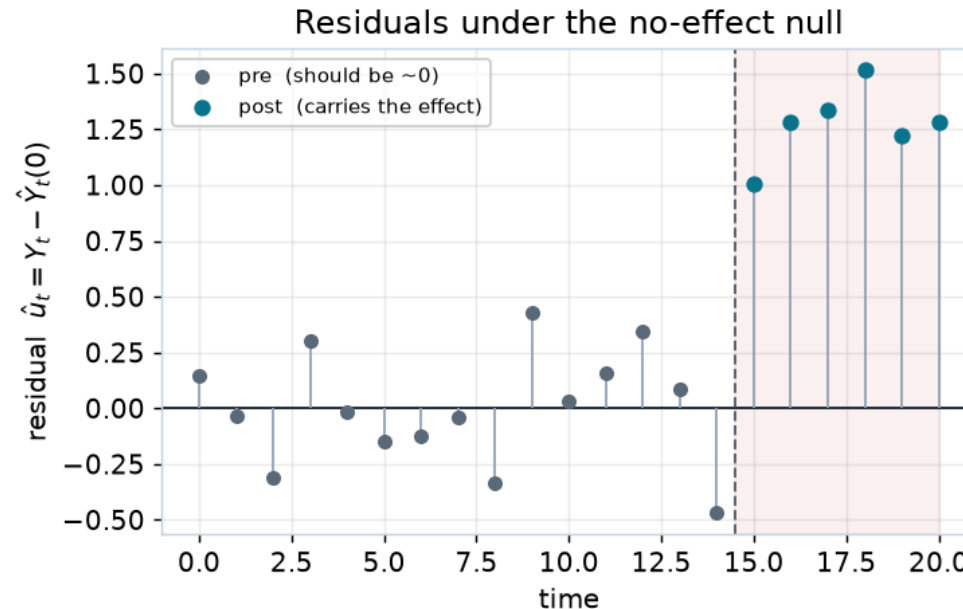
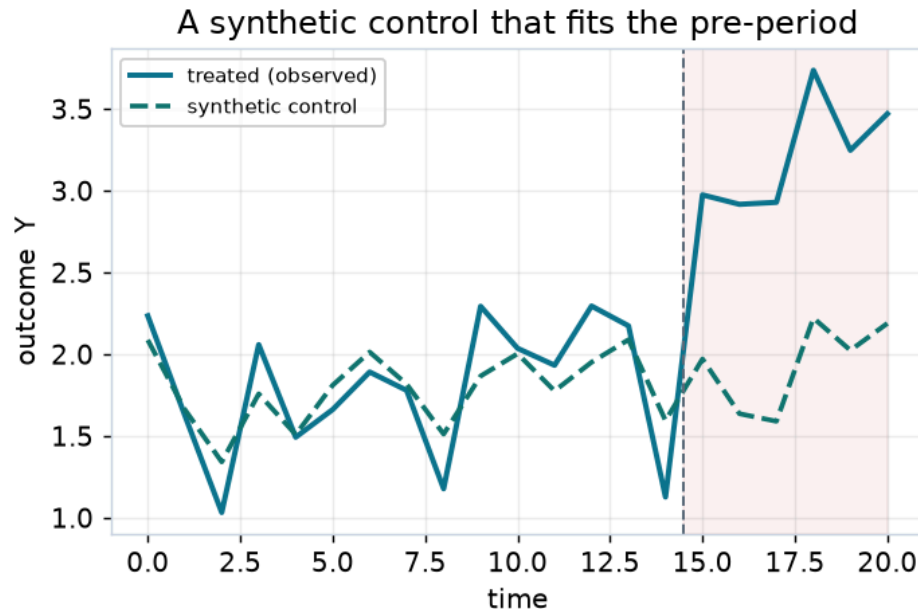
Setup — we predict the counterfactual

- Before the test, we need the ingredients: a treated unit, a donor pool, and a prediction of what would've happened without treatment — nothing new if you sat through the Panel Models deck.
- One treated unit Y_{1t} , a donor pool Y_{0t} , pre-periods $t \leq T_0$ and post-periods $t > T_0$. Fit a **synthetic control** $\hat{Y}_{1t}(0)$ (from the Panel Models deck).
- The **gap** $\hat{u}_t = Y_{1t} - \hat{Y}_{1t}(0)$ is the residual. Under *no effect*, that residual process has **no break** at T_0 .
- The residuals over time are serially dependent — so this is exactly the non-exchangeable case, handled by **moving-block** permutations from the last section.

Step 1 — impose the null, read the residuals

- We're testing a hypothesized effect size θ — a sharp null on the post-period effect, $H_0 : \tau_t = \theta$. Subtract it from the **post outcomes only**, refit over the **whole timeline**, and take residuals for every t :

$$\tilde{Y}_{1t} = Y_{1t} - \theta \cdot \mathbf{1}\{t > T_0\}, \quad \hat{u}_t = \tilde{Y}_{1t} - \hat{Y}_{1t}(0)$$



One Number, Tested Against Its Shuffled Twins

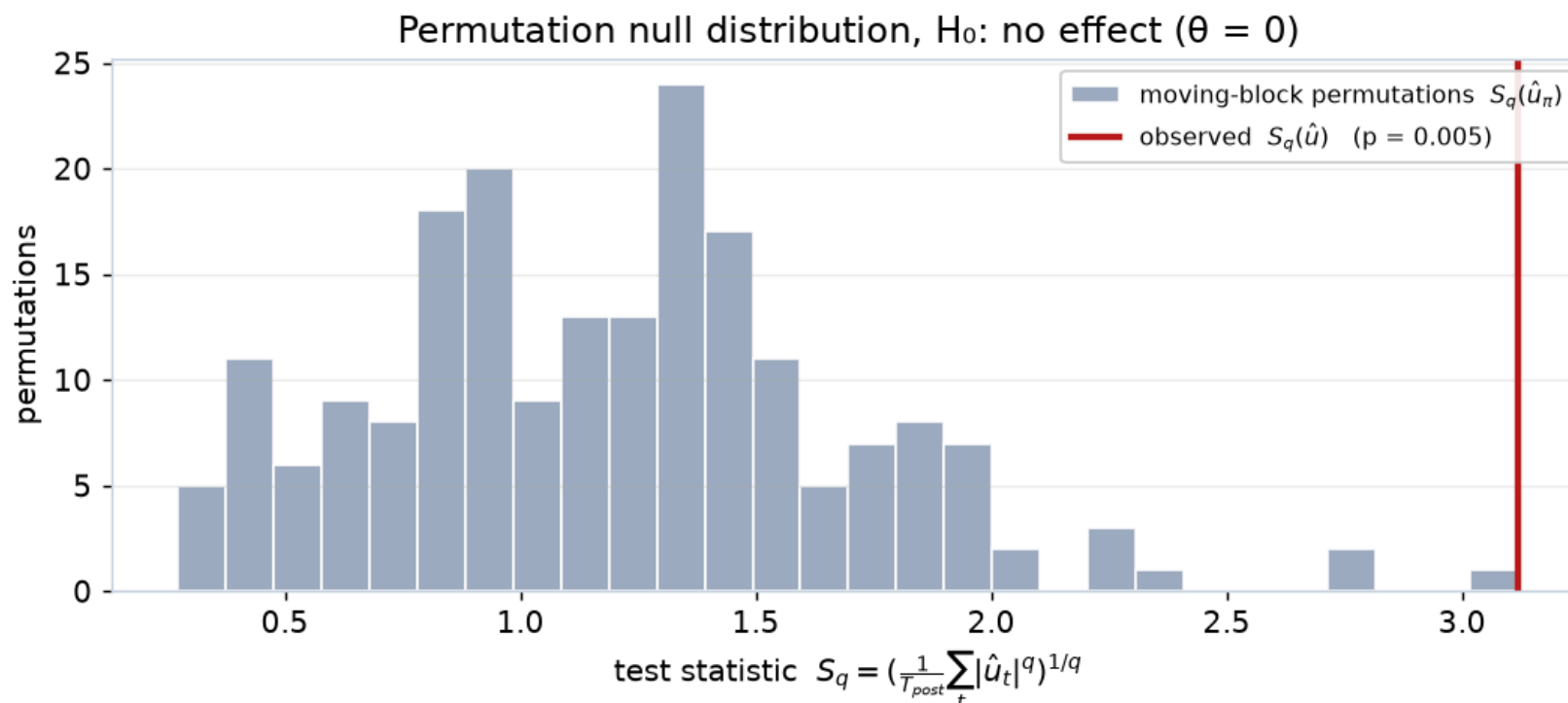
- Summarize the post-period residuals with a single number ($q = 1$ = mean absolute residual), so "the residuals look big" becomes something you can actually test:

$$S_q(\hat{u}) = \left(\frac{1}{T_{post}} \sum_{t > T_0} |\hat{u}_t|^q \right)^{1/q}$$

- Compare S_q to its distribution under **moving-block permutations** of the residual sequence — the serial-dependence-robust scheme from the last section.
- `panelib`'s `conformal_inf` implements exactly this, so you don't have to.
- Details: [Appendix — Choosing the Test Statistic](#)*

Step 3 — the permutation p-value

$$p = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S_q(\hat{u}_\pi) \geq S_q(\hat{u})\}, \quad \text{reject } H_0 \text{ if } p \leq \alpha.$$

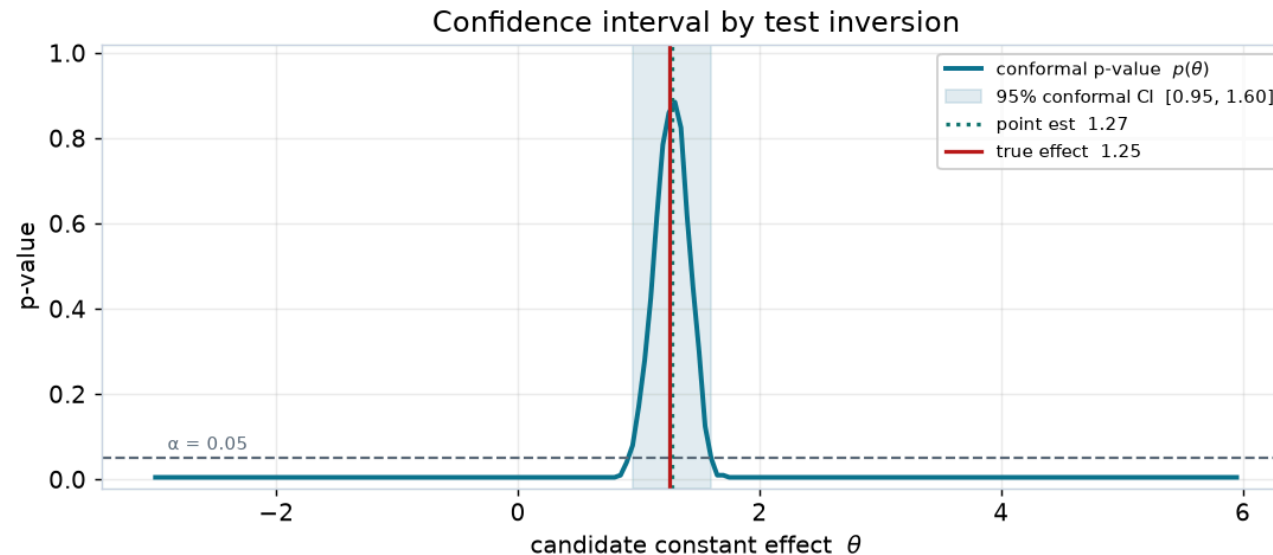


- **Careful:** this permutes residuals **across time** for one treated unit — not the Abadie placebo test, which permutes treatment **across units**.

Step 4 — confidence interval by test inversion

- Repeat the test over a grid of candidate effects θ and keep every value you **cannot reject** — that's a full confidence interval, not just a yes/no test:

$$CI_{1-\alpha} = \{ \theta : p(\theta) > \alpha \}$$



- The point estimate sits where $p(\theta)$ peaks; the CI is the set of effects consistent with "no break."

Why this matters

- **The payoff, one more time:** this works with **one treated unit and few pre-periods** — and is **exact** under exchangeability, **robust** to serial dependence and to a **misspecified** counterfactual (validity comes from residual symmetry, not a correct model).
- The fine print: under dependence the guarantee is approximate, and it needs the counterfactual's **error to be stable over time** — a model whose bias *trends* into the post-period breaks a symmetry no permutation can restore.
- Resolution matters too: with very few post-periods the permutation p-value is coarse — in the Panel deck's simulations (2 post-periods) coverage ran **80–85%**, not 95%.
- Contrast Abadie placebo inference: its p-value resolution is only $1/(J + 1)$ and it needs a large donor pool.
- This is the inference the **Panel Models** deck used to put DiD, SC, and SDID on the same footing.

Conclusion

- **Back to the big idea:** conformal inference buys an **honest, finite-sample guarantee** from **one assumption you can actually reason about** — exchangeability — and it degrades gracefully, rather than catastrophically, when that assumption is stressed.
- Reach for it to **wrap black-box predictors** you don't fully trust, and — the reason we built this whole deck — to do inference where classical SEs go silent: **a single treated unit over time**.
- The price: coverage is **marginal**, not conditional; and under dependence you must **block** or **weight** to keep the guarantee.
- Bottom line: start from the guarantee you can actually defend, then pick the score and the permutation that fit your data.

Appendix

Appendix: split vs. full conformal

- **Split (inductive)** conformal — the recipe here — spends part of the data on a calibration fold. Cheap (one fit), but the interval is random in that split and a bit wider.
- **Full** conformal refits for every candidate label — exact but expensive; **CV+ / jackknife+** (Barber et al. 2021) recover most of the efficiency without a held-out split.
- Same guarantee; the choice is **data efficiency vs. compute**.

Appendix: the coverage law

- Conditional on the calibration draw, the coverage of split conformal is a **random variable**:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C} \mid \text{calibration}) \sim \text{Beta}(k, n + 1 - k), \quad k = \lceil (n + 1)(1 - \alpha) \rceil.$$

- Its mean is $\geq 1 - \alpha$; the spread shrinks like $1/\sqrt{n}$ — the middle histogram made this concrete.
- The upper bound $1 - \alpha + \frac{1}{n+1}$ needs the scores to have **no ties** (continuous scores, or a random tie-break).

Appendix: Choosing a Conformity Score

- Any score $s(x, y)$ is valid for the guarantee — the only thing at stake in the choice is **efficiency** (set tightness), never validity.
- Common families beyond the absolute residual $|y - \hat{f}(x)|$:
 - **Normalized:** $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$ — divides by a local scale so the score reflects *relative* surprise (the fix behind **conformalized quantile regression**, Romano, Patterson & Candès 2019).
 - **Quantile-based:** $s = \max\{\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)\}$ — built from two fitted quantile regressions.
 - **Classification margin:** $s = 1 - \hat{\pi}_y(x)$ — the standard score outside regression.
- Rule of thumb: pick the score that captures *where* your model is uncertain — a score blind to heteroskedasticity gives valid but needlessly wide sets exactly where the model is confident.

Appendix: Why Coverage Survives a Bad Model

- Split conformal's guarantee has two separable pieces: **validity** (coverage $\geq 1 - \alpha$) and **efficiency** (set tightness). The rank argument on "Why it works" only ever uses exchangeability — $\hat{f}$ never enters it.
- Extreme case: let $\hat{f}(x)$ be the training fold's unconditional mean, ignoring x entirely. The set is far wider than a well-fit model's — but it still covers $1 - \alpha$ of the time, since calibration recenters around whatever the score distribution actually is.
- This is why conformal is a popular *wrapper* around models you don't fully trust — validity doesn't require the wrapped model to be well specified, only that the calibration fold is exchangeable with the test point.
- Back to [*The Guarantee Holds Even When Your Model Is Wrong*](#)

Appendix: Conformal Sets as Test Inversion

- $p(y)$ is a genuine p-value: under exchangeability and $H_0 : Y_{n+1} = y$, $p(y)$ is (super-)uniform on $[0, 1]$ — the same guarantee classical hypothesis tests offer, with no distributional assumption beyond exchangeability.
- This makes conformal prediction a special case of **Neyman's construction**: build a valid test for every candidate value, then invert by collecting every value not rejected. Classical intervals from pivotal statistics are the parametric special case; conformal drops the pivot requirement entirely.
- The duality runs both ways: any exchangeability-respecting permutation test induces a valid conformal-style set by the same inversion — exactly the move the synthetic-control section makes. The permutation p-value there **is** this $p(y)$, with y replaced by a hypothesized treatment effect θ .
- Back to [*Conformal Prediction Is Hypothesis Testing in Disguise*](#)

Appendix: another route — prediction intervals

- Conformal is not the only distribution-aware option for synthetic control.
- **Cattaneo, Feng & Titiunik (2021)** build **prediction intervals** that separate two error sources: the in-sample **weight-construction** error and the out-of-sample **stochastic** error, with finite-sample guarantees.
- Rule of thumb: **conformal** tests a sharp null by permutation and inverts to a CI; **prediction intervals** model the two error sources directly. Different assumptions, complementary diagnostics.

Appendix: conformal for treatment effects, beyond panels

- **Individual treatment effects** — Lei & Candès (2021): conformal intervals for $Y(1) - Y(0)$ under unconfoundedness, using *weighted* conformal to bridge the treated/control covariate shift. The natural companion to the HTE deck.
- **Covariate shift** — Tibshirani, Barber, Candès, Ramdas (2019): weighted conformal stays valid when the test point comes from a different covariate distribution with a known (or estimated) likelihood ratio.
- **Drift in production** — Gibbs & Candès (2021): *adaptive* conformal updates the level α_t online to keep realized coverage on target under distribution shift — the monitoring use case for deployed models.

Appendix: The CWZ Framework — Assumptions

- Formally, CWZ needs the post-period residuals to be (approximately) **exchangeable with the pre-period residuals under H_0** — delivered by **stationarity** of the residual process plus **weak dependence** (mixing decays fast enough that a block permutation approximately preserves the joint law).
- The framework is agnostic to *how* the counterfactual is built: DiD, synthetic control (ADH), synthetic DiD, or a factor model all plug in as the same $\hat{Y}_{1t}(0)$ — only the residual's exchangeability under the null matters, not the estimator's asymptotics.
- **Validity vs. power** are cleanly separated: a poor counterfactual still yields a valid test (residuals stay symmetric under H_0), but a good one yields a *powerful* test — model quality shows up in power, never in size.
- *Continued next slide — comparisons and the multi-unit case.*

Appendix: The CWZ Framework — Comparisons

- **Abadie placebo** permutes treatment across *units* and needs a sizable donor pool for p-value resolution; **CWZ** permutes residuals across *time* for a single unit and needs enough pre-periods instead. Different resampling axis, same test-inversion logic.
- For **multiple treated units**, see the Panel Models deck's appendix on inference with many treated units — CWZ's single-unit permutation isn't the right tool once you have several.
- Back to *Causal Inference = Prediction + a Break Test*

Appendix: Choosing the Test Statistic

The summary statistic $\mathcal{S}_q$ trades off what kind of break it's sensitive to:

- $q = 1$ (mean absolute) — the deck's default; balanced power against a sustained, moderate shift.
- $q = 2$ (root-mean-square) — more sensitive to a single large spike in the post-period than to many small ones.
- $q = \infty$ (max absolute deviation) — detects a single sharp break, at the cost of power against a gradual trend.
- *Continued next slide — block length and the permutation count.*

Appendix: Choosing the Permutation Scheme

- **Block length** trades bias against variance: too short and the permutation shatters real local dependence (inflated size); too long and too few distinct block placements are left to permute (coarse, low-power p-values). Heuristic: set it near the residual series's estimated dependence range.
- **Number of permutations $|\Pi|$** : exhaustive block permutations are cheap for short pre-periods; for longer series, Monte Carlo sampling from Π gives the same test with a controllable p-value resolution ($1/|\Pi|$ sets the finest achievable p-value).
- This is exactly what `panelib`'s `conformal_inf` exposes as tunable arguments — the defaults above are its choices.
- *Back to [One Number, Tested Against Its Shuffled Twins](#)*

Literature Review of Related Papers

- **Conformal prediction — foundations**

- Vovk, Gammerman, Shafer (2005). *Algorithmic Learning in a Random World*. Springer — the founding monograph. [link](#)
- Shafer & Vovk (2008). "A Tutorial on Conformal Prediction." *JMLR* 9, 371–421. [link](#)
- Lei, G'Sell, Rinaldo, Tibshirani, Wasserman (2018). "Distribution-Free Predictive Inference for Regression." *JASA* 113(523). [link](#)
- Angelopoulos & Bates (2023). "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." *Found. & Trends ML*. [link](#)

Literature Review (continued)

- **Adaptivity & beyond exchangeability**

- Romano, Patterson, Candès (2019). "Conformalized Quantile Regression." *NeurIPS*. [link](#)
- Barber, Candès, Ramdas, Tibshirani (2021). "Predictive Inference with the Jackknife+." *Ann. Statist.* 49(1). [link](#)
- Barber, Candès, Ramdas, Tibshirani (2023). "Conformal Prediction Beyond Exchangeability." *Ann. Statist.* 51(2), 816–845. [link](#)
- Tibshirani, Barber, Candès, Ramdas (2019). "Conformal Prediction Under Covariate Shift." *NeurIPS*. [link](#)
- Gibbs & Candès (2021). "Adaptive Conformal Inference Under Distribution Shift." *NeurIPS*. [link](#)

Literature Review (continued)

- **Conformal inference for causal panels / synthetic control**

- Chernozhukov, Wüthrich, Zhu (2018). "Conformal Inference Methods for Predictive ML with Dependent Data." *COLT (PMLR 75)*. [link](#)
- Chernozhukov, Wüthrich, Zhu (2021). "Conformal Inference for Counterfactual and Synthetic Controls." *JASA* 116(536). [link](#)
- Cattaneo, Feng, Titiunik (2021). "Prediction Intervals for Synthetic Control Methods." *JASA* 116(536). [link](#)
- Firpo & Possebom (2018). "Synthetic Control Method: Inference, Sensitivity Analysis and Confidence Sets." *J. Causal Inference* 6(2). [link](#)
- Lei & Candès (2021). "Conformal Inference of Counterfactuals and Individual Treatment Effects." *JRSS-B* 83(5). [link](#)

- **Companion code:** `panelib.py` (`conformal_inf`) in [statanomics](#).

